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Question

Calculate the mass of 1 atom of nitrogen.

Solution

1. The molar mass of a compound defines the mass of 1 mole of that substance.
2. We can also say the gram atomic mass is the total mass of all the atoms in grams that make a mole of atoms.

Given:

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Now,

$$5. \text{ Mass of } 6.022 \times 10^{23} \text{ atoms} = 14\text{g}$$

$$6. \text{ Mass of 1 atom} = \frac{14}{6.022 \times 10^{23}}\text{g}$$

$$7. \text{ Mass of 1 atom} = 2.32 \times 10^{-23}\text{g}$$

Final Answer

The mass of 1 atom of Nitrogen is $2.32 \times 10^{-23}\text{g}$.

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Q. Calculate the mass of 1 atom of nitrogen.

Q. calculate the mass of -1). 1 mole of nitrogen atoms {At.mass of nitrogen = 14u}2]. 2 mole of aluminium atoms [At.mass of al = 27u).

Q. Calculate the mass of 3.011310×10^{23} atom of nitrogen(Atomic mass of N=14u)

Q. Calculate the mass of 1 molecule of Nitrogen (N_2). (Atomic mass of N = 14).

Q. Calculate the mass of 3.011×10^{23} number of nitrogen atoms.

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Stoichiometric Calculations

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